

Ideation of Solution Directions

This section presents the ideation stage of the project. The purpose of this stage is to generate and explain multiple distinct concept directions that could address the solution specifications identified in GP2. These concepts are not final solutions. Instead, they represent different ways the problem could be approached, each based on a different perspective such as governance, process, structure, technology, or human behavior.

In GP2, the core issue was not the absence of research activity, but the absence of a consistent system that makes research transparent, traceable, verifiable, and reusable over time. This means the solution must do more than improve one isolated part of the process. It must respond to the full set of specifications derived from GP2, including documentation clarity, decision traceability, structured storage, process visibility, and continuity between project cycles.

Each concept is grounded in academic research and institutional practice. This ensures that the ideas are not based on assumptions, but on established knowledge and proven practices. The ideas below are therefore early concept directions that show different ways the VCH Lab could respond to the same underlying problem. Each idea is explained as a complete approach in its own right so that it can later be compared, validated with stakeholders, and either refined, rejected, or combined into a stronger final solution.

To ensure full clarity, each concept is structured around three core questions: what the idea is, what purpose it serves within the VCH Lab, and how it would work from start to finish in practice. This makes the ideation stage not only exploratory, but also directly useful as the foundation for selecting and developing the final GP3 solution.

Solution concepts

1. Governance-based research policy

This idea treats the VCH problem primarily as a lack of clear rules, responsibilities, and accountability. At the moment, students are expected to produce research outputs, but no sufficiently formal system defines exactly what must be documented, who is responsible for doing so, and what minimum standards must be met before work can be considered complete. As a result, documentation quality depends too much on individual judgment, motivation, and interpretation. This leads to inconsistency across projects and weak continuity between student teams.

The idea itself is to create a formal governance policy for research documentation and reproducibility within the VCH Lab. In simple terms, this would be a lab level policy that defines the minimum requirements for how research must be recorded and managed. It would not just say that documentation is important. It would specify what kinds of documentation are

mandatory, such as source records, decision logs, metadata, validation notes, and final handover materials. It would also define who is responsible for maintaining those records during a project and what standards must be met before a project is archived or passed on.

The purpose this serves within VCH is to create a shared baseline. Instead of each team deciding for itself what is enough, the lab would have a consistent set of expectations. This would support students because they would know from the beginning what is required. It would support supervisors because they would have a clearer basis for monitoring project quality. It would also support the lab as a whole because archived projects would follow the same minimum structure, making them easier to compare, validate, and reuse.

From start to finish, this idea would work as follows. At the start of every project, students and supervisors would receive the governance policy as part of the onboarding and project briefing. The policy would define the required documentation outputs for the project and explain the minimum standards for traceability and storage. During the project, students would be expected to follow these rules as part of their regular work, not only at the end. Supervisors would use the same policy as a reference point during feedback and review moments. Before a project can be considered complete, a final check would confirm whether the required documentation standards have actually been met. Once approved, the project would be archived according to the same governance requirements, ensuring that it is not only finished but also reusable.

This idea is supported by the broader logic of structured research stewardship in the FAIR principles (Wilkinson et al., 2016), by the OECD emphasis on institutional responsibility for research data access and management (OECD, 2021), by Science Europe's guidance on aligned research data management requirements (Science Europe, 2022), and by the University of Twente's policy approach to research data management (University of Twente, 2023).

This directly addresses the GP2 finding that documentation practices in VCH are inconsistent and depend too heavily on individual student interpretation rather than a shared standard.

2. Stage gate research lifecycle model

This idea treats the VCH problem primarily as a process and timing issue. In GP2, one of the clearest findings was that documentation usually happens too late. Final outputs are often present, but the intermediate steps that explain how the research developed are missing. This means that important parts of the process, such as shifts in scope, key decisions, analysis steps, and validation moments, are not captured when they happen. Once those moments pass, they are difficult or impossible to reconstruct accurately.

The idea itself is to redesign the VCH research process into a stage gate model. A stage gate model means that the project is divided into clearly defined phases, and each phase has required outputs that must be completed before the next phase can begin. In the VCH context, this could include stages such as defining the problem, collecting data, analysing findings,

validating ideas, developing the solution, and preparing the final handover. At each stage, specific documentation must be completed and reviewed before the project moves on.

The purpose this serves within VCH is to make documentation part of the process itself instead of something added later. This would help students record their work at the moment it actually happens, rather than trying to reconstruct it afterward. It would also help supervisors because they would be able to check progress and quality at logical moments in the research lifecycle. Most importantly, it would make the project process visible from beginning to end, which is exactly what is currently missing.

From start to finish, this idea would work like this. At the beginning of the project, the research would be divided into a fixed number of stages. For each stage, the required documentation and outputs would already be defined. For example, the first stage might require a defined problem statement, stakeholder overview, and initial research direction. The next stage might require documented sources, interview planning, and clear raw data records. Later stages might require structured analysis notes, decision justifications, or validation outcomes. Before the project can move from one stage to the next, these outputs would be checked. If they are incomplete, the team would need to complete them first. By the end of the project, this would create a full chain of visible steps, rather than a final report with missing middle stages.

This idea is supported by Science Europe's lifecycle based approach to research data management (Science Europe, 2022), by the University of Twente's emphasis on data management across all stages of research (University of Twente, 2023), and by the iterative, staged logic of Design Science Research (Hevner et al., 2004).

This directly addresses the GP2 finding that documentation in VCH is concentrated near the submission stage and that intermediate steps are often not preserved.

3. Standardized documentation template system

This idea treats the VCH problem primarily as a documentation consistency issue. Even when students do document parts of their work, they often do so in different formats, with different levels of detail, and with different assumptions about what matters. This makes projects difficult to compare and even harder to understand later. One team may include detailed interview notes, while another includes only a final summary. One student may explain why a decision was made, while another only records the final result. This inconsistency weakens traceability.

The idea itself is to create a standardized system of templates for recording research work. These templates would not replace the research itself. Instead, they would give students a clear and repeatable format for capturing key parts of the process. This could include templates for interview documentation, source tracking, analysis notes, decision logs, validation records, and handover summaries. Each template would guide the user to record the right information in the right place.

The purpose this serves within VCH is to create consistency without requiring every student to invent their own documentation structure. It helps ensure that important information is captured in comparable ways across projects. This would make it easier for supervisors to review projects, for future students to understand earlier work, and for the lab to build a more reusable body of research over time.

From start to finish, this idea would work like this. At the start of a project, students would receive a template pack that corresponds to the main research activities they are expected to carry out. As they conduct interviews, gather sources, analyse information, make decisions, and validate ideas, they would fill in the relevant templates. These templates would guide them to record not only the output, but also the reasoning, assumptions, and evidence behind it. During the project, supervisors could check whether the templates are being used consistently. At the end, the completed templates would form part of the project archive, making the research process much easier to follow.

This idea is supported by the FAIR emphasis on contextual and structured information for reuse (Wilkinson et al., 2016), by Science Europe's alignment of documentation expectations (Science Europe, 2022), and by the OECD emphasis on preserving not just outputs but also the materials needed for interpretation and reuse (OECD, 2021).

This directly addresses the GP2 finding that reasoning, intermediate analysis, and decision trails are often missing or unclear even when final outputs exist.

4. Repository and metadata architecture redesign

This idea treats the VCH problem primarily as a storage and retrieval issue. GP2 showed that many materials are stored somewhere, but not in a way that supports interpretation, navigation, or reuse. Files are fragmented, naming is inconsistent, relationships between documents are unclear, and important contextual information is missing. This means that even when data exists, it is often not independently understandable to someone who was not part of the original team.

The idea itself is to redesign the VCH repository and its metadata structure. A repository is the place where project materials are stored. Metadata is the structured information that explains what a file is, when it was created, by whom, what it contains, and how it relates to the rest of the project. Together, repository structure and metadata determine whether stored work can actually be found, understood, and reused.

The purpose this serves within VCH is to transform the repository from a passive storage location into an organized research environment. Instead of just collecting final files, the repository would show the logic of the project. A user could move through the research in a structured way, understand what each file is for, and see how raw materials connect to later analysis and outputs.

From start to finish, this idea would work as follows. A standard folder structure would be created for all VCH projects, based on the project lifecycle. Each folder would correspond to a stage of the research process, such as setup, data collection, analysis, validation, and final outputs. Within each folder, files would follow naming conventions so that their purpose is immediately clear. Each important file would also include metadata, either built into the system or recorded through a structured file description. This would make it possible to understand not only where a file is stored, but also what role it plays in the research process. At the end of the project, the archive would already be structured for reuse rather than needing to be reorganized later.

This idea is supported by the FAIR principles on findability and reuse (Wilkinson et al., 2016), by SURF Yoda as a lifecycle based RDM environment (SURF, 2023), by 4TU.ResearchData as a structured research repository model (4TU.ResearchData, 2023), and by University of Twente guidance on FAIR data and preservation (University of Twente, 2023).

This directly addresses the GP2 finding that VCH storage is fragmented, inconsistent, and difficult to interpret without prior project involvement.

5. Digital traceability platform or dashboard

This idea treats the VCH problem primarily as one of visibility and system support. In GP2, one of the recurring problems was that there is no clear, central place where the progress of the research process can be followed. Data, decisions, validation, and outputs often exist in separate places, which makes it difficult to see how the project developed over time.

The idea itself is to create a digital workspace, platform, or dashboard that actively supports the research process. This would not just be another storage place. It would be an environment where project stages are visible, decisions are logged, evidence is linked, and progress can be tracked. The purpose of the platform would be to make the process visible while it is happening, rather than only after the fact.

The purpose this serves within VCH is to provide a stronger operational layer for traceability. Instead of relying on scattered files and memory, students and supervisors would be able to see the current state of the project, what has already been documented, what is still missing, and how decisions connect to the evidence collected. This could also support supervision by making research progress more transparent in real time.

From start to finish, this idea would work like this. At the start of the project, a digital workspace would be set up for the team. As the project moves forward, each stage would be recorded in the system. Students would upload data, connect files, record key decisions, and update progress. Validation moments could also be recorded, creating a visible trail of feedback and changes. Supervisors could review the workspace during the project instead of waiting for final

outputs. At the end, the platform would contain a traceable record of the project process, not just a final report.

This idea is supported by the OECD's broader view that research outputs include workflows, software, and intermediate materials, not only final data products (OECD, 2021), by FAIR's emphasis on connected and reusable information structures (Wilkinson et al., 2016), and by SURF Yoda's lifecycle support model (SURF, 2023).

This directly addresses the GP2 finding that the relationship between data, decisions, and outputs is often invisible and not centrally traceable.

6. Training and behavioral adoption model

This idea treats the VCH problem primarily as a behavior and adoption issue. Even if strong systems, templates, and repository structures are introduced, they will not work properly if students and supervisors do not understand why they matter or how they should be used. GP2 showed that documentation is often seen as secondary to the final deliverable, and that many important steps remain tacit because they are not recognized as something that should be recorded.

The idea itself is to focus on training, onboarding, guidance, and reinforcement. This means helping students and supervisors understand the purpose of reproducibility, what good documentation looks like, and how to apply the expected practices consistently throughout the project.

The purpose this serves within VCH is to support the human side of change. Structural improvements alone may fail if people do not adopt them. This idea aims to strengthen awareness, discipline, and consistency so that reproducible practices become part of normal project work rather than feeling like extra effort.

From start to finish, this idea would work like this. At the beginning of the semester, students and supervisors would receive onboarding on the purpose of reproducibility and the documentation standards expected in VCH. Examples of strong and weak practice could be shown so expectations become concrete. During the project, regular reminders, support moments, and feedback would reinforce these practices. Supervisors would also be equipped to model and monitor the expected behavior. By the end of the project, students would not only have produced better documentation, but also developed stronger research habits that could continue into later work.

This idea is supported by 4TU.ResearchData's combination of repository services and training support (4TU.ResearchData, 2023), by University of Twente's service and support model (University of Twente, 2023), and by broader RDM guidance that assumes early support and planning are necessary for quality practice (Science Europe, 2022; OECD, 2021).

This directly addresses the GP2 finding that documentation often depends on informal habits, personal judgment, and unclear expectations rather than stable routines.

7. Integrated socio technical reproducibility framework

This idea treats the VCH problem as a system level problem that cannot be solved by one intervention alone. A socio technical problem means that the issue exists both in how people behave and in how the surrounding systems, rules, and tools are designed. In VCH, reproducibility does not fail for only one reason. It fails because governance, process, storage, documentation, and user behavior are disconnected from each other.

The idea itself is to create one integrated reproducibility framework that combines the strongest elements of the previous ideas into a single coherent way of working. This means using governance rules to define expectations, lifecycle stages to define timing, templates to define how work is recorded, repository structure to define where information is stored, and validation checkpoints to ensure the quality and visibility of the process.

The purpose this serves within VCH is to solve the full problem rather than only one part of it. It recognizes that policy alone is too abstract, workflow alone is too narrow, storage alone is insufficient, and training alone is too weak. What VCH needs is an approach where these elements support each other and function as one connected system.

From start to finish, this idea would work like this. At the start of a project, students and supervisors would receive a clear framework that defines expectations, stages, tools, and responsibilities. The project would then move through a structured lifecycle with required checkpoints and documentation outputs at each stage. Students would use standardized templates to record their work, and all files would be stored in a repository that follows a clear architecture and metadata logic. Supervisors would review progress at key validation moments. By the end of the project, the result would not only be a final deliverable, but a complete, traceable, and reusable record of how the research was conducted and why decisions were made.

This idea is strongly aligned with Design Science Research because it focuses on developing a practical solution to a real organizational problem through iteration, testing, and refinement (Hevner et al., 2004). It is also consistent with the combined logic of FAIR, OECD, Science Europe, and Dutch RDM practice, all of which show that reproducibility depends on the alignment of rules, process, infrastructure, and user behavior (Wilkinson et al., 2016; OECD, 2021; Science Europe, 2022; SURF, 2023; 4TU.ResearchData, 2023; University of Twente, 2023).

This directly addresses the GP2 finding that the main issue in VCH is not the absence of isolated components, but the lack of a connected system that makes research work transparent, traceable, and reusable from start to finish.



Concept Comparison Table (with Academic Grounding)

Idea	Core Problem Framing	What the Solution Does	Strengths	Limitations	Academic / Institutional Grounding
1. Governance based reproducibility policy	Lack of accountability and clear rules	Defines mandatory documentation standards, roles, and responsibilities across projects	Creates clarity, consistency, and enforceability; aligns with institutional practice	Does not ensure execution in practice; depends on compliance	FAIR Principles (Wilkinson et al., 2016); OECD (2021) Recommendation on Access to Research Data; Science Europe RDM Guide (2022); University of Twente RDM Policy
2. Stage gate research lifecycle model	Documentation not embedded in process timing	Introduces mandatory checkpoints per project phase where documentation must be completed before progressing	Ensures documentation happens at the right moment; improves traceability across lifecycle	Does not define <i>how</i> documentation is structured or stored	Science Europe lifecycle approach (2022); University of Twente lifecycle RDM guidance; Design Science Research iterative structuring (Hevner et al., 2004)
3. Standardized documentation template system	Inconsistent and incomplete documentation practices	Provides structured templates for data capture, decision logs, validation, and reporting	Improves consistency and comparability across projects; easy to implement	Does not ensure usage or integration into workflow; can become superficial if not enforced	FAIR Principles (findability, interoperability); OECD (2021); Science Europe (2022) emphasis on contextual metadata and documentation

4. Repository and metadata architecture redesign	Poor storage, structure, and retrievability of research outputs	Creates structured repository with lifecycle folders, metadata standards, and clear file relationships	Enables long term reuse and independent understanding; improves accessibility	Does not guarantee that correct data is entered or maintained	FAIR Principles; SURF Yoda infrastructure; 4TU.ResearchData repository model; University of Twente FAIR data guidance
5. Digital traceability platform or dashboard	Lack of visibility and linkage between research steps	Provides a system where decisions, data, and progress are tracked and linked dynamically	Enhances transparency and real time tracking; supports workflow execution	Requires technical development and adoption; may be complex for users	OECD (2021) expanded scope including workflows and software; SURF Yoda lifecycle environment; FAIR machine readable logic
6. Training and behavioral adoption model	Weak adoption and inconsistent research behavior	Focuses on onboarding, guidance, and reinforcing good documentation practices	Supports cultural change and improves user compliance	Weak if implemented alone; does not structurally enforce behavior	4TU.ResearchData training and community support; University of Twente RDM services; Science Europe and OECD emphasis on early planning and support structures
7. Integrated socio technical reproducibility framework	Misalignment between people, processes, and systems	Combines governance, lifecycle checkpoints, templates, repository structure, and validation into one coherent framework	Addresses full system; ensures alignment across all dimensions; strongest long term solution	More complex to design and implement; requires coordination across elements	Design Science Research (Hevner et al., 2004); FAIR; OECD (2021); Science Europe (2022); SURF Yoda; 4TU.ResearchData; University of Twente

How you use this in GP3 (VERY IMPORTANT)

This table is not just descriptive — it is your **argument tool**.

◆ Step 1 — Show breadth (Sufficient)

You demonstrate:

- multiple concepts
 - different perspectives
 - coverage of the full problem space
-

◆ Step 2 — Show comparison (Good)

Use the table to explicitly state:



- each idea solves a different part of the problem
 - no single idea is sufficient on its own
-

◆ Step 3 — Show selection logic (Excellent)

You MUST explicitly write:



“Based on stakeholder validation, the following concepts were identified as the strongest: governance, lifecycle, repository, and integrated system alignment.”

Then:



“These concepts were selected because they address complementary dimensions of the problem.”

◆ Step 4 — Show rejection (CRUCIAL FOR EXCELLENT)

You must say:



- templates alone → too limited
- training alone → not structural
- digital platform alone → too dependent on adoption

◆ Step 5 — Lead into Idea 7



“This comparison revealed that no single concept could fully address the problem, leading to the development of an integrated framework.”

Why this table is strong

It proves:

- ✓ multiple perspectives
- ✓ structured thinking
- ✓ academic grounding
- ✓ justification of choices
- ✓ logical transition to final solution